

PCA

PRINCIPAL COMPONENT ANALYSIS (PCA)

→ PCA is a statistical technique that transforms correlated variables into a smaller set of uncorrelated variables called Principal Components while preserving maximum information from the original dataset.

- Dimensionality Reduction
- Feature Extraction
- Data Compression
- Pattern Recognition
- Noise Reduction

→ Why PCA could be important in this era

- Modern systems generate massive amount of data such as

Field	Data Generated
Healthcare	MRI Scans, Sensors
Climate Science	Satellite data
Smart Cities	Traffic & Pollution Sensors
Online Banking	Transactions
etc.	

How PCA could help us:

→ Imagine a classroom dataset containing:

- Study Hours
- Attendance
- Quiz Marks
- Assignment Marks
- Final Exam Marks

Most of these variables are related.

For example:

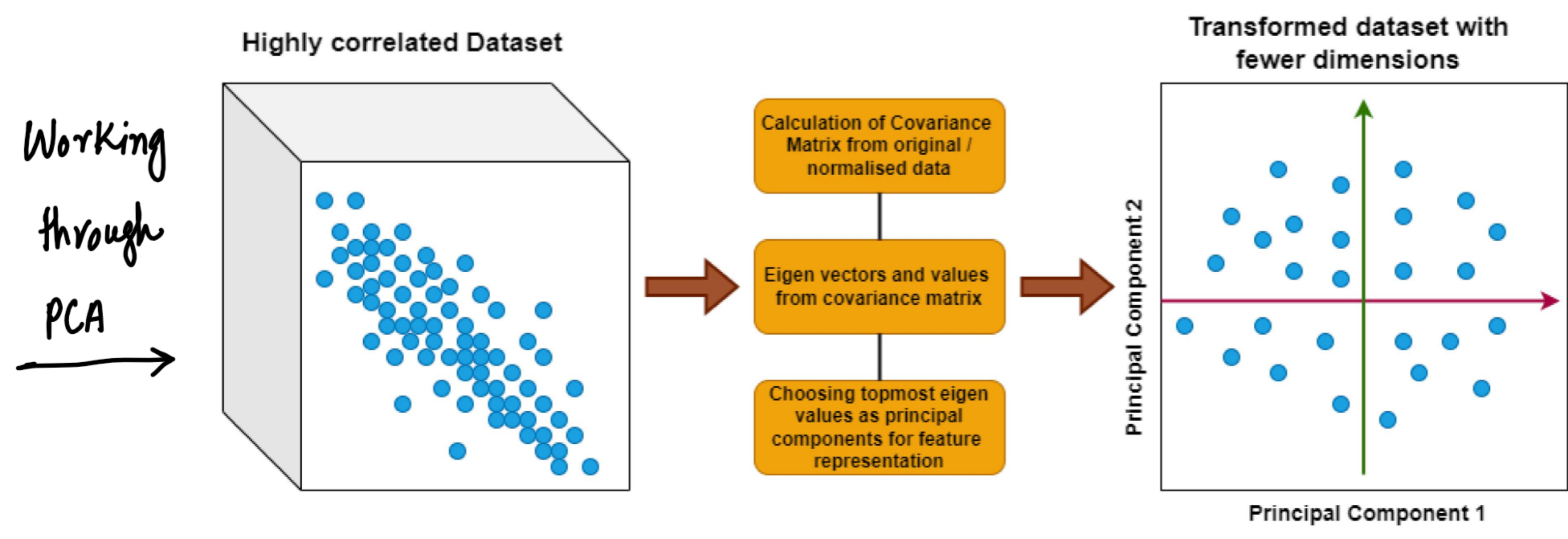
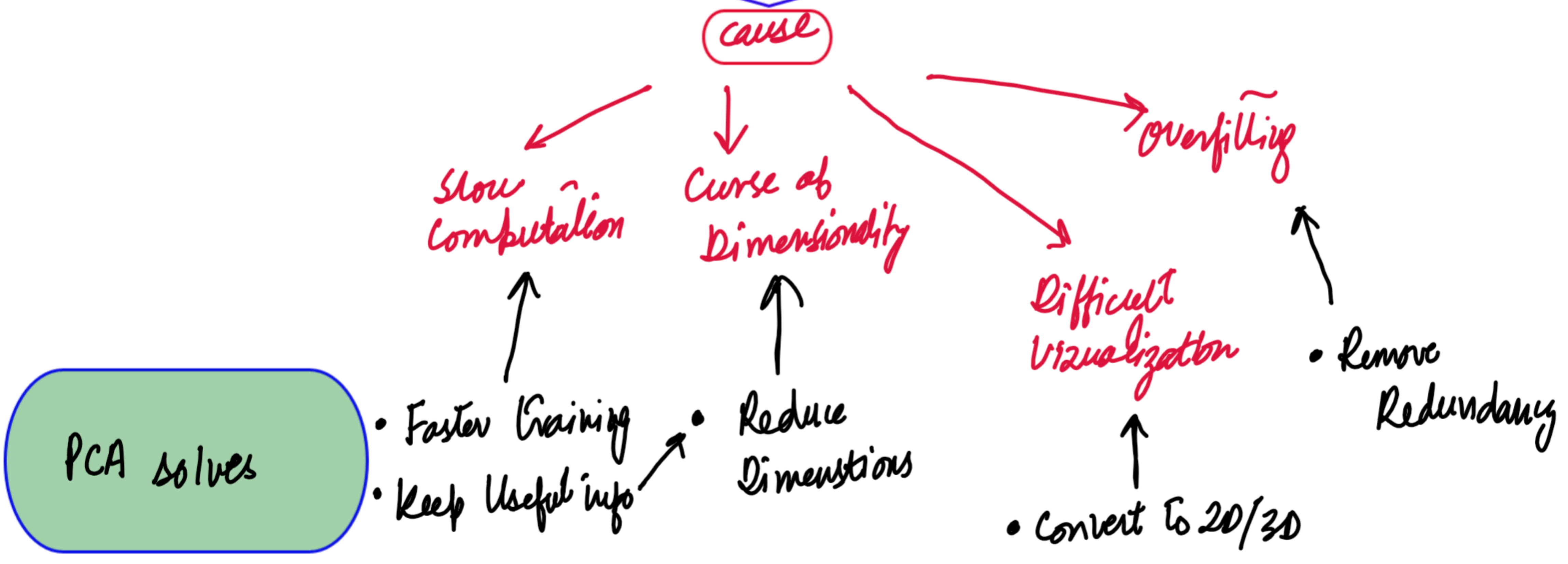
Students who study more usually get higher marks.
Students with better attendance often perform better.

Therefore, many variables contain similar information.

PCA helps us:

- Remove redundancy
- Keep only the most important information
- Reduce complexity

→ Suppose a dataset has 100 variables.
"Do we really need 100 all?
Maybe only → 5 variable contain most information"



PCA MATHEMATICAL FOUNDATION FOR PCA

1. STANDARDIZATION:

$$z = \frac{x - \mu}{\sigma}$$

2. COVARIANCE MATRIX:

- Tells how variables move together

$$\bar{X} = \frac{\sum X}{n}; \bar{Y} = \frac{\sum Y}{n}$$

$$\text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n-1}$$

- Covariance Matrix Structure

(For two variables)
etc (X_1, X_2)

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_1, X_2) & \text{Var}(X_2) \end{bmatrix}$$

$$\text{Var}(X_1) = \frac{\sum (x_{1i} - \bar{x}_1)^2}{n-1}$$

$$\text{Cov}(X_1, X_2) = \frac{\sum (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{n-1}$$

* Characteristic Eq

$$|A - \lambda I| = 0 \rightarrow \text{Eigen Value}$$

3. EIGEN VALUES AND EIGEN VECTORS:

(represents magnitude of variance)

(direction of maximum variance)

Covariance
Matrix

$$Av = \lambda v$$

$$\begin{cases} Av - \lambda v = 0 \\ v(A - \lambda I) = 0 \end{cases}$$

Eigen Vector

- Large Eigenvalue $\rightarrow$ Important Comp.
- Small " $\rightarrow$ less " "

4. PRINCIPAL COMPONENT

$\rightarrow$ Ranked based $\rightarrow$

PC1

PC2

PC3

$\rightarrow$ contains max. information from data.

5. PROJECTION

$\rightarrow$ Original data is projected into principal components.

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EXAMPLE : AIR POLLUTION ANALYSIS IN SMART CITIES

A government wants to analyze pollution in smart cities using vehicle density and CO₂ emissions rate (data provided in Table below). Use a suitable ML-algo to help policy makers in identifying main pollution factor and also helps in reducing computational aspects in regards to variables.

City	Vehicle Density	CO ₂ Emission
A	20	30
B	25	35
C	40	50
D	50	65

x_1

x_2

$n = 4$

SOLUTION:

Step-I: Compute Mean:

$$\bar{x}_1 =$$

$$\bar{x}_2 =$$

Step-II: Compute Variance:

$$\text{Var}(x_1) =$$

$$\text{Var}(x_2) =$$

Step-III: Compute Covariance:

$$\text{Cov}(x_1, x_2)$$

Step-III(a): Covariance Matrix

$$\bar{X} = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_1, x_2) & \text{Var}(x_2) \end{bmatrix} = \begin{bmatrix} & \\ & \end{bmatrix}$$

x_1	x_2	$x_{1i} - \bar{x}_1$	$x_{2i} - \bar{x}_2$	$(x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)$
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Step - IV: Compute Eigenvalues =

Solve:

$$|\bar{X} - \lambda I| = 0$$

Identity Matrix i.e. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ • magnitude of
variance

$$\left| \bar{X} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0 \rightarrow \left| \bar{X} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

→ apply determinant
→ use quadratic eq.
for λ_1, λ_2

→ PC1 → λ_1 PC2 → λ_2 Explained Variance
 $PC_1 = \frac{\text{component Var}(\lambda_1)}{\text{Total Var}}$ $PC_2 = \frac{\text{component Var}(\lambda_2)}{\text{Total Var}}$ Step - V: Compute Eigenvectordirection
of maximum
variance

$$(A - \lambda I)v = 0$$

PC1 eigenvalue

Assume $v = \begin{bmatrix} x \\ y \end{bmatrix}$

$$(A - \lambda_1 I) \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

→ use simultaneous equation
to solve for eigen vector

→ PCA usually uses normalized
vector

• Find magnitude first

$$m = \sqrt{x^2 + y^2}$$

then

$$v = \begin{bmatrix} x/m \\ y/m \end{bmatrix}$$

→ Normalized Vector

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EXAMPLE: MENTAL HEALTH AND DIGITAL LIFESTYLE

A university is studying the impact of digital lifestyle on student mental health. The following data was collected from students: Daily Screen Time (hours) & Stress Level Score. The dataset is given as

Student	Screen Time (X_1)	Stress Score (X_2)
A	3	40
B	5	50
C	7	65
D	9	80

Using Principal Component Analysis (PCA), analyze the dataset and interpret the results. Your analysis should include necessary calculations, principal components, explained variance, and interpretation of the hidden learning pattern discovered through PCA.